Experiment No.	03
Experiment Title	Regression Analysis using Linear and Regularized Models
GitHub Repository:	https://github.com/KJayasuriya/ml_laboratory
Student Name	K. Jayasuriya
Register Number	3122247001026
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

To implement linear and regularized regression models for predicting a continuous target variable, evaluate their performance using multiple metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2 Problem Statement

A real-world regression dataset containing numerical and categorical features related to loan appli- cations is used. The target variable is the loan amount sanctioned.

- Implement, Linear, Ridge, Lasso, ElasticNet regression models.
- Tune regularization hyperparameters using Grid Search or Randomized search.
- Visualize regression results and errors.
- Analyze overfitting, underfitting and bias-variance trade-off.

3 Dataset Description

Dataset Name	Loan Prediction
Dataset Source	https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset
Number of Samples	614
Number of Features	12
Missing Values	122
Train-Test Split	Train-80% Test-20%

4 Exploratory Data Analysis

Statistical summary

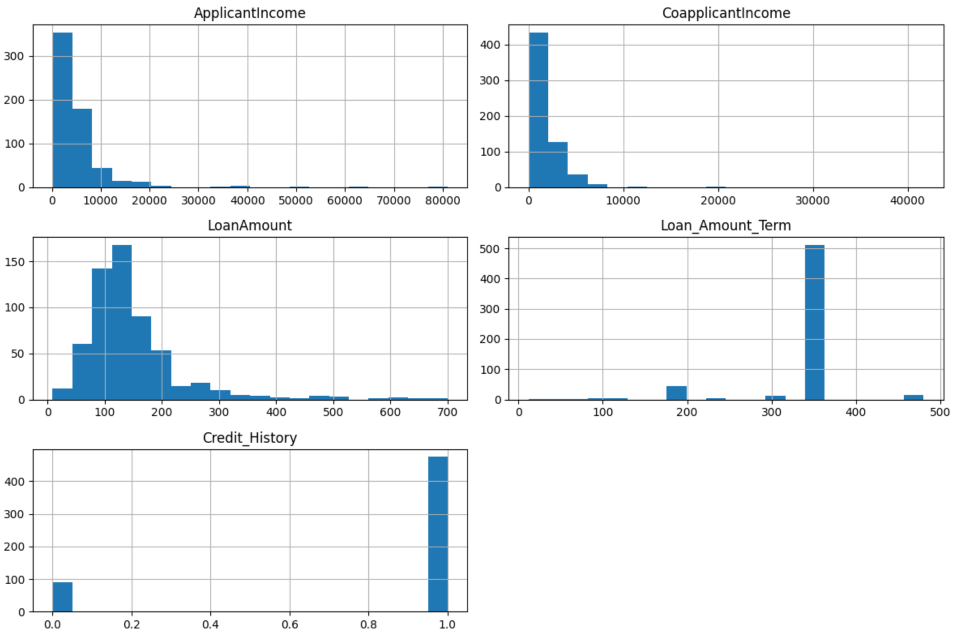


Figure 1: Histograms showing Statistical summary

Inference:
This histogram shows, among the numerical columns credit History is only feature having 2 distinct values so it is a categorical column with number datatype. and LoanAmount target feature is distributed normally.

Correlation of the dataset features

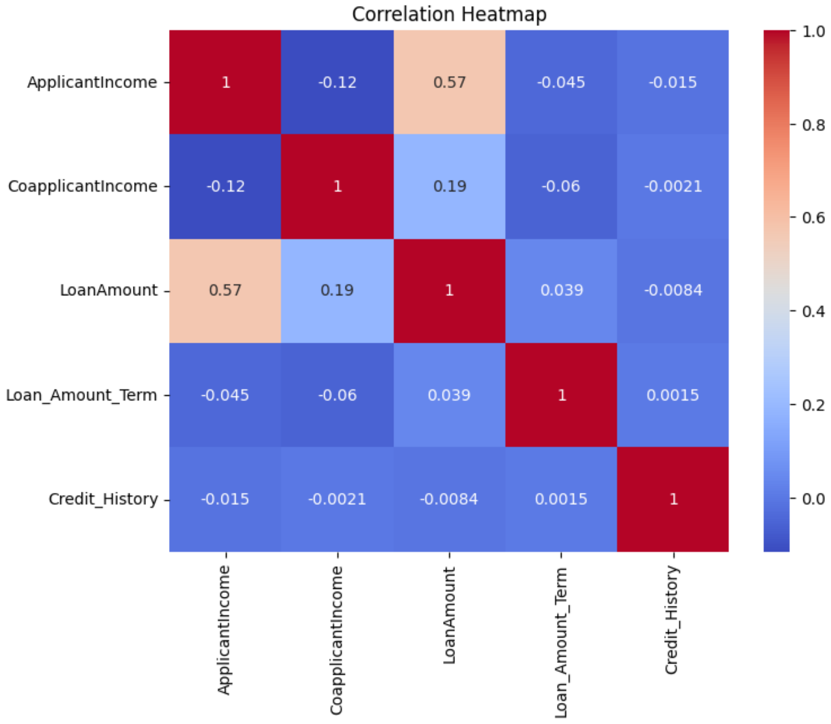


Figure 2: Correlation of features

Inference:
See the correlation between ApplicantIncome and LoanAmount it is 0.57. So the heatmap tells LoanAmount is highly correlated by Applicant Income. As income increases Loan amount also increases (Directly proportional).

5 Data Preprocessing

5.1 Missing Value Handling:

Missing values in numerical attributes were replaced using the median value, while missing values in categorical attributes were replaced using the most frequent category using the `SimpleImputer` class.

5.2 Encoding:

Categorical variables such as Gender, Married, Education, Self_Employed and Property_Area were converted into numerical format using One-Hot Encoding to make them suitable for machine learning algorithms.

5.3 Feature Scaling:

Numerical attributes were standardized using `StandardScaler` so that all features have zero mean and unit variance. This improves model stability during training.

5.4 Train-Test Split:

The dataset was divided into training and testing sets using an 80:20 ratio with a fixed random state for reproducibility.

5.5 Feature Engineering:

No additional feature engineering techniques were applied. The original dataset features were directly used after preprocessing.

6 Machine Learning Algorithm

6.1 Working Principle

Linear Regression is a supervised learning algorithm used to predict continuous numerical values. It models the relationship between the dependent variable and one or more independent variables by fitting the best straight line that minimizes prediction error.

6.2 Mathematical Formulation

The prediction equation is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where

- y is the predicted output,
- β_0 is the intercept,
- β_i are regression coefficients,
- x_i are input features.

The model minimizes the Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

6.3 Advantages

- Simple and easy to interpret.
- Fast to train.
- Works well when the relationship between variables is approximately linear.
- Computationally efficient.

6.4 Limitations

- Assumes a linear relationship.
- Sensitive to outliers.
- Cannot capture complex nonlinear relationships.
- Performance decreases when assumptions are violated.

7 Experimental Setup

Python Version	3.13.13
Scikit-Learn Version	1.7.5
IDE	Jupyter Notebook
Operating System	Fedora Linux 42
Hardware	Intel core i5 (11th gen) 8GB RAM

8 Hyperparameter Selection

Parameter	Value
Search Method	GridSearchCV
Cross-validation	5-fold
Ridge alpha	100
Lasso Alpha	10
ElasticNet alpha	1
ElasticNet L1 Ratio	0.5

9 Results

Linear Regression

Metric	Value
Mean Absolute Error	36.988
Mean Squared Error	3339.922
Root Mean squared Error	57.792
R2 score	0.094

Ridge regression

Metric	Value
Mean Absolute Error	36.971
Mean Squared Error	3334.977
Root Mean Squared Error	57.749
R2 score	0.095

Lasso regression

Metric	Value
Mean Absolute Error	35.841
Mean Squared Error	3232.042
Root Mean Squared Error	56.851
R2 score	0.123

ElasticNet regression

Metric	Value
Mean Absolute Error	37.826
Mean Squared Error	3048.685
Root Mean Squared Error	55.215
R2 score	0.173

Inference:
Among 4 regression models, Lasso Regression model has the lowest MAE and ElasticNet regression has the lowest RMSE.

9.1 Best Model

Model	Best CV R^2
Linear Regression	0.1840
Lasso	0.2544
ElasticNet	0.3384
Ridge	0.3407

Since Ridge regression has highest Best CV it performs best on this dataset.

9.2 Bias Variance Trade-off

We know that, a model is said to be **Overfitting** if it has **Low Bias and High variance** or Good Train accuracy and bad test accuracy else if model has both low train and test accuracies or **High bias and Low variance** it is called as **Underfitting Model**.

From the output:

Model	Train score	Test Score	Outcome
Linear Regression	0.4567	0.0939	Underfitting
Lasso	0.45123	0.1232	Underfitting
ElasticNet	0.3928	0.1729	Underfitting
Ridge	0.4567	0.0953	Underfitting

Inference: The training and testing scores of all four regression models are relatively low, indicating that the models are unable to capture the underlying relationship between the input features and the target variable effectively. Although there is a difference between the training and testing scores, the training performance is not sufficiently high to classify the models as overfitting. Hence, the models exhibit underfitting (high bias). Among the evaluated models, ElasticNet achieved the highest testing score, while Ridge Regression obtained the best cross-validation performance after hyperparameter tuning.

10 Result Visualization

Actual vs Predicted results

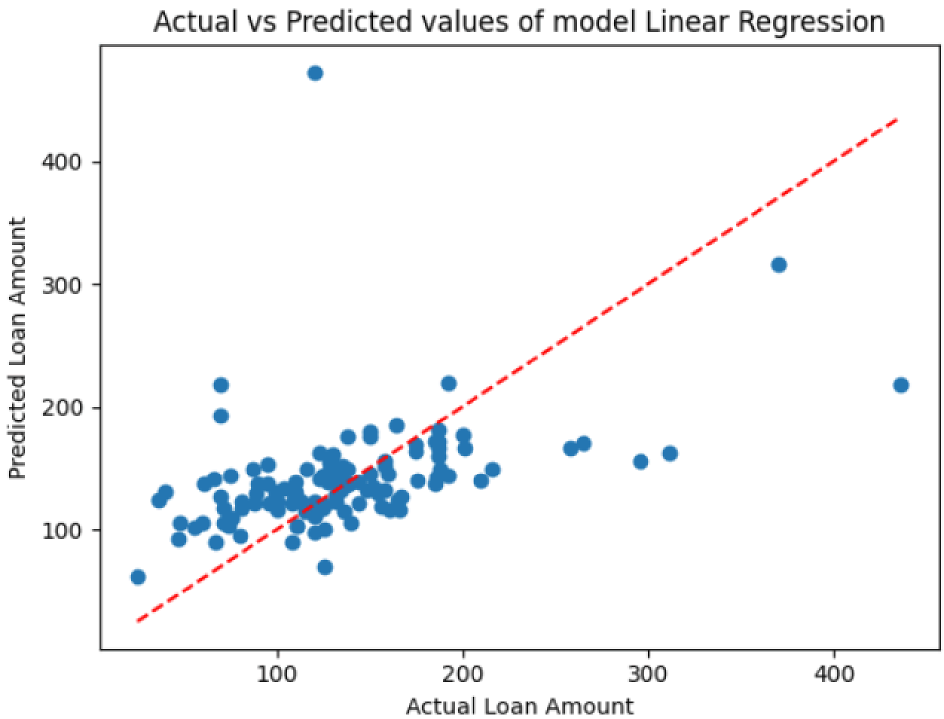
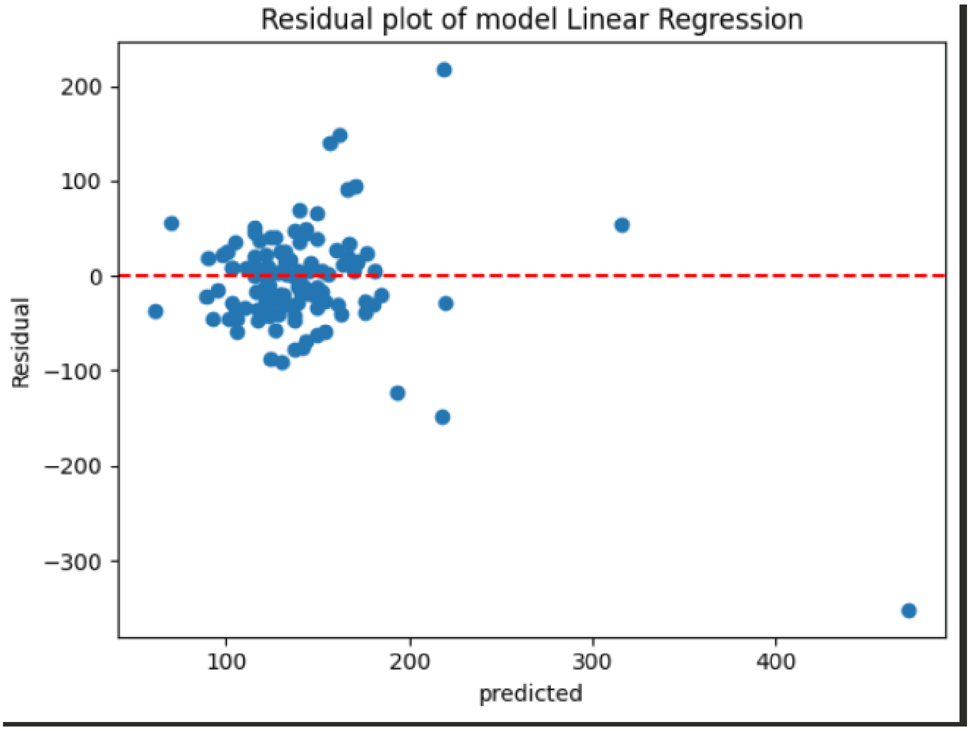


Figure 3: Actual VS Predicted value plot

Inference:
x-axis as actual value (True value) and y-axis as Predicted value. The imaginary red line passing diagonally is the perfect prediction line which is used to compare our model.

Residual Plot



Inference: This is like normalized version of previous graph. where, x-axis is predicted y value. y-axis is the residual value i.e. difference between actual y and predicted y values. The horizontal red line the "zero residual" model.

11 Discussion

Trained all 4 variants of regression. But the limitation is before calculating performance metrics and plotting visualization plots it trains the model for all the 4 algorithms and predicts that is computationally expensive.

12 Conclusion

From this experiment, I have learnt to Train a model with LinearRegression, Lasso, Ridge and ElasticNet regression and computed their performance using metrics such as, MAE, MSE, RMSE, R2 score. Also performed Cross validation.

13 References

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